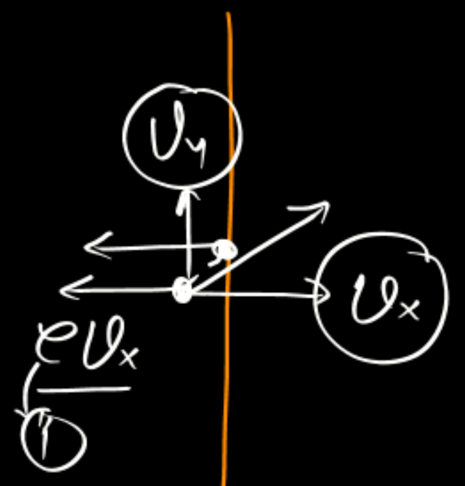
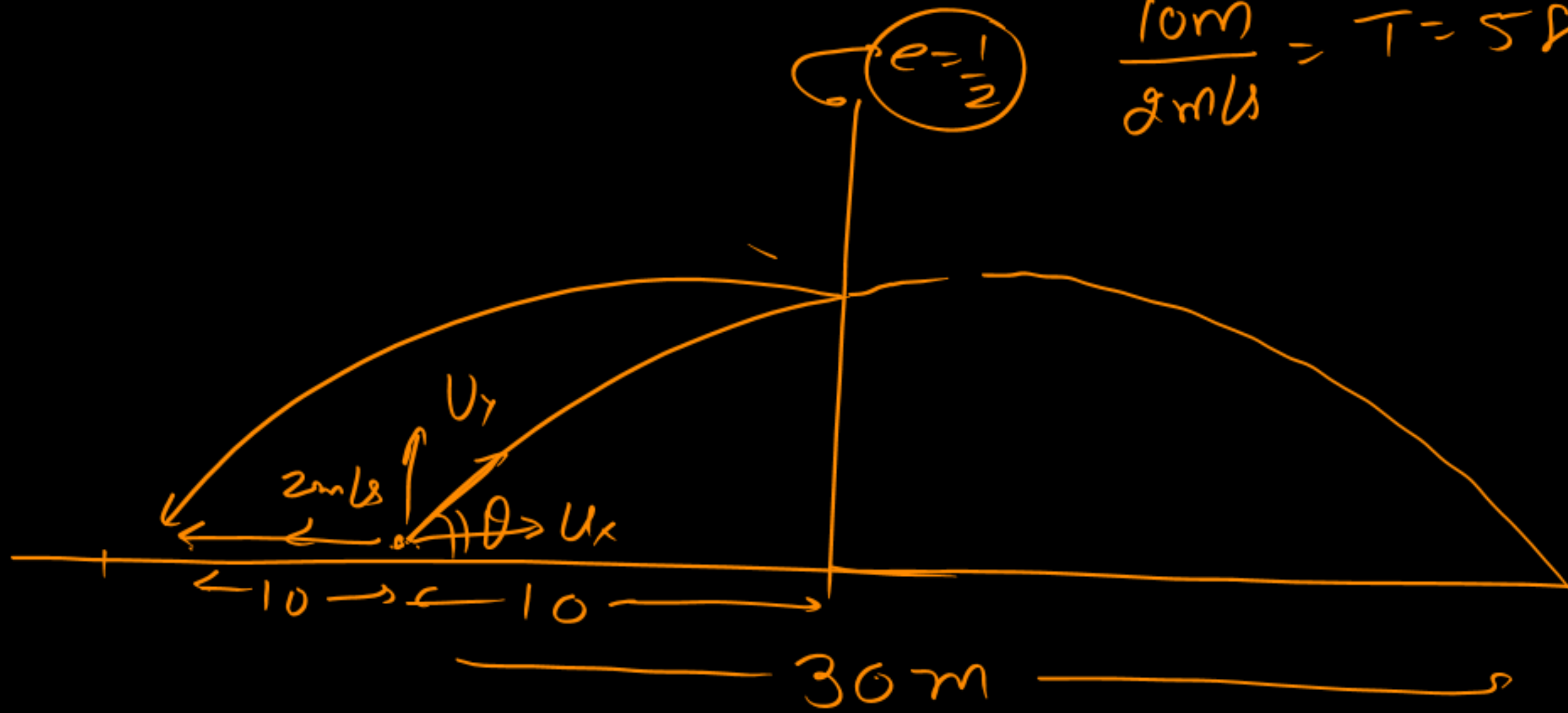
A student is standing infront of a very high smooth vertical wall at a perpendicular separation 10m. He throws a ball towards wall and to catch the same ball he run with constant speed 2 m/s opposite to wall just after throw it. If he catches the ball after running 10 m distance. Then projection angle is (assume collision of ball with wall is elastic) : (use g = 10 m/s^2)

SCQ

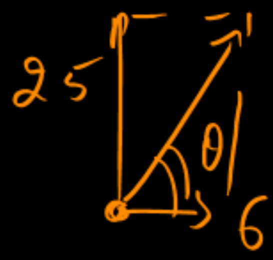
(A) $\theta = \tan^{-1}\left(\frac{25}{6}\right)$ (B) $\theta = \tan^{-1}\left(\frac{30}{8}\right)$ (C) $\theta = \tan^{-1}\left(\frac{15}{6}\right)$ (D) $\theta = \tan^{-1}\left(\frac{10}{6}\right)$





$\frac{10m}{2m/s} = T = 5\,sec = \frac{2U_y}{10}$

$\boxed{U_y = 25\,m/s}$



$U = \sqrt{25^2 + 6^2}$

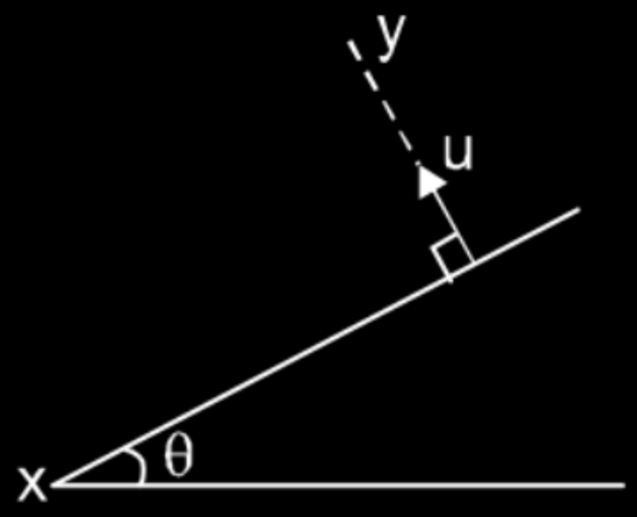
$\tan\theta = \frac{25}{6}$

$30 = \frac{2U_y U_x}{10}$

$300 = 2 \times 25\, U_x$

$U_x = 6$

A ball is projected perpendicularly from an inclined plane of angle θ, with speed 'u' as shown. The time after which the projectile is making angle 45º with the inclined plane is :



(A) $\dfrac{u}{g\sin\theta}$ (B) $\dfrac{u}{g\cos\theta}$ (C) $\dfrac{u}{g\{\sin\theta+\cos\theta\}}$ (D) $\dfrac{u}{g\{\sin\theta-\cos\theta\}}$

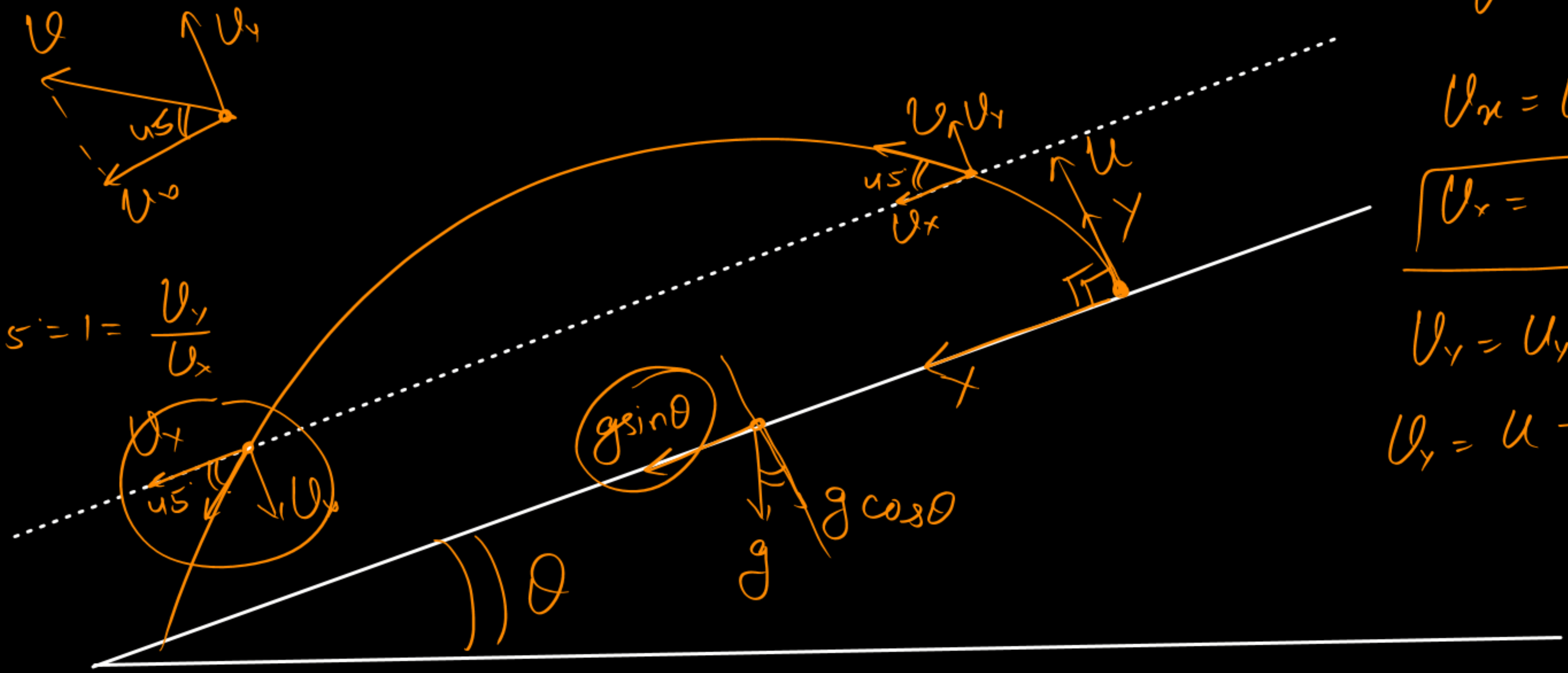
$v = u + at$

$v_x = u_x + a_x t$

$\boxed{v_x = 0 + g\sin\theta\, t}$

$v_y = u_y + a_y t$

$v_y = u - g\cos\theta\, t$



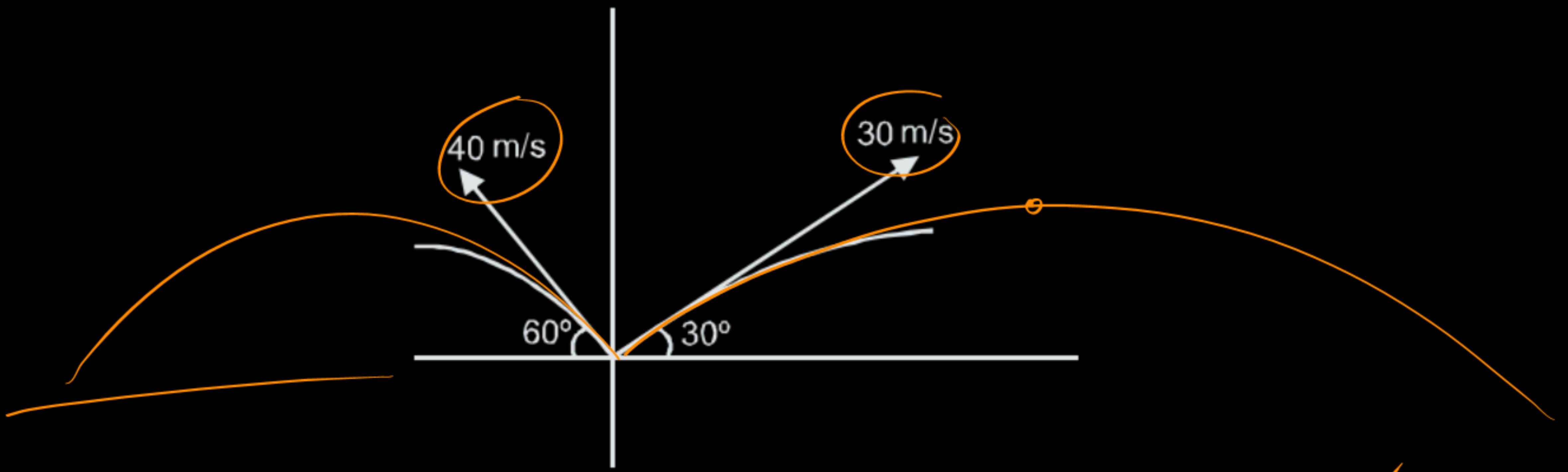
$\tan 45 = 1 = \frac{v_y}{v_x}$

$v_y = v_x$

$u - g\cos\theta . t = g\sin\theta\, t \Rightarrow t = \frac{u}{g\{\sin\theta + \cos\theta\}}$

Two particles A & B are thrown from same point and in same vertical plane as shown in figure with velocity 30 m/s & 40 m/s. Find the separation between them after 3 sec.

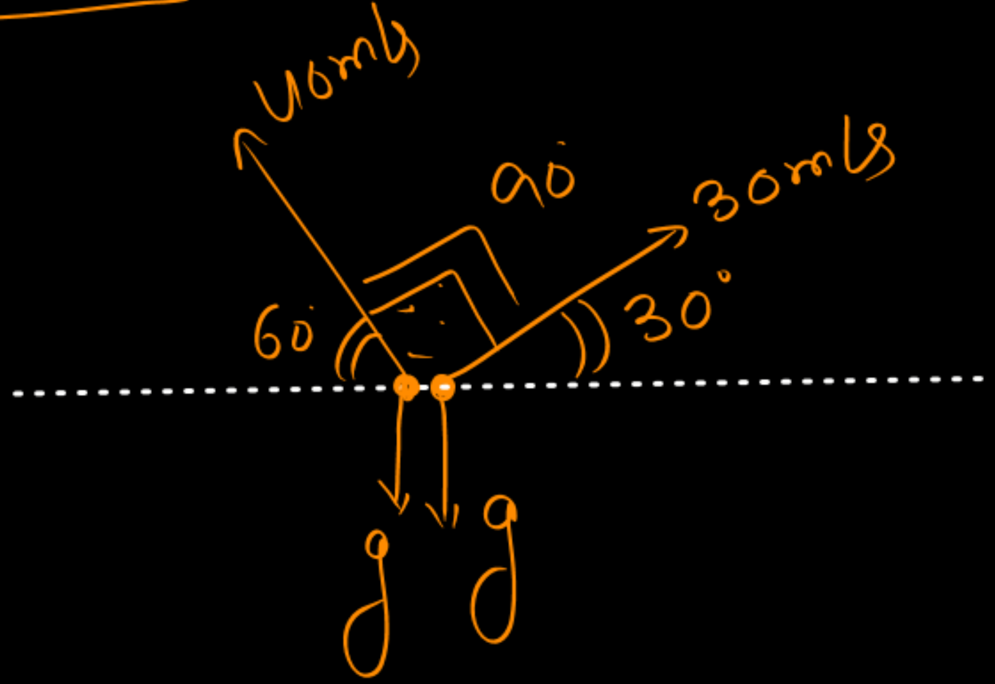
{SC8}



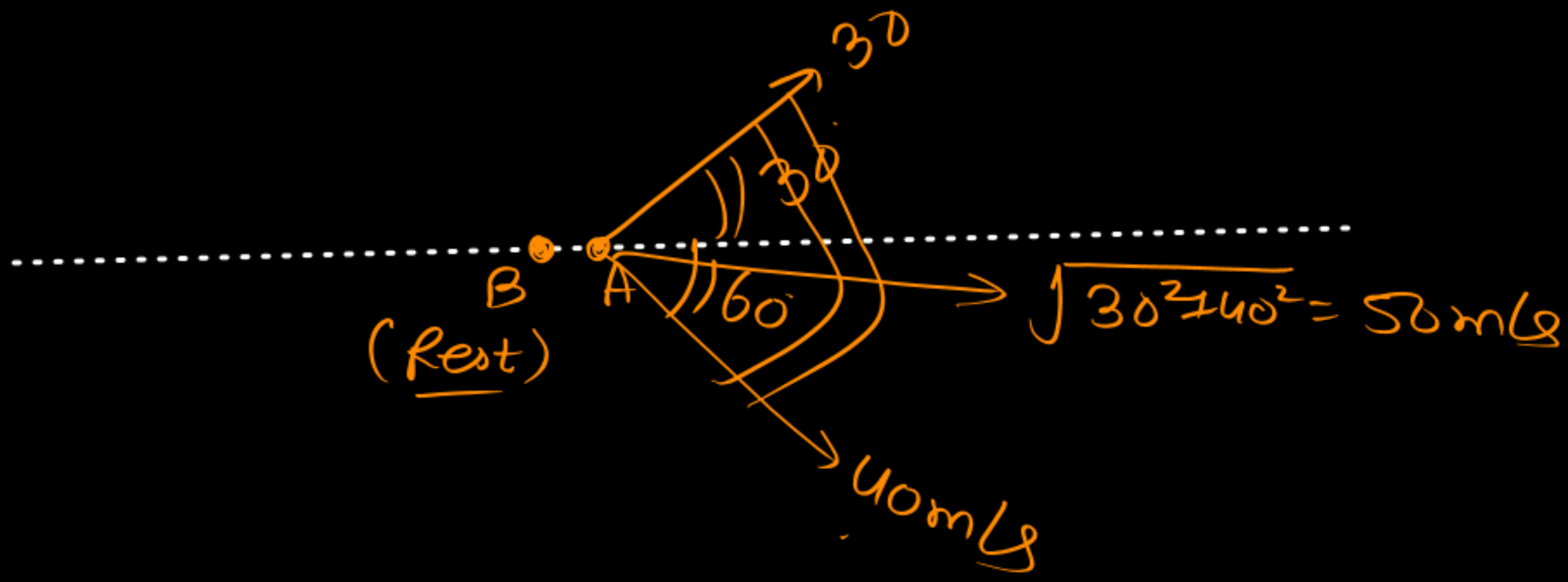
(A) 100 m (B) 115 m (C) 250 m (D) 150 m

wrt B.

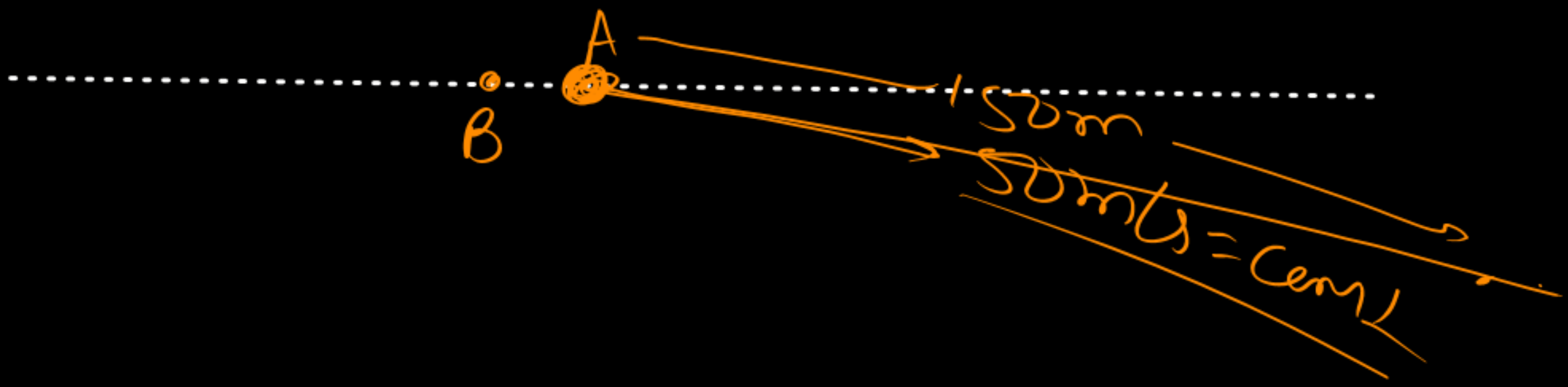
$a_{AB} = 0$



40m/s
90°
30m/s
30°
60°
g
g



30
30°
B
A
(Rest)
60°
$\sqrt{30^2+40^2} = 50$m/s
40m/s

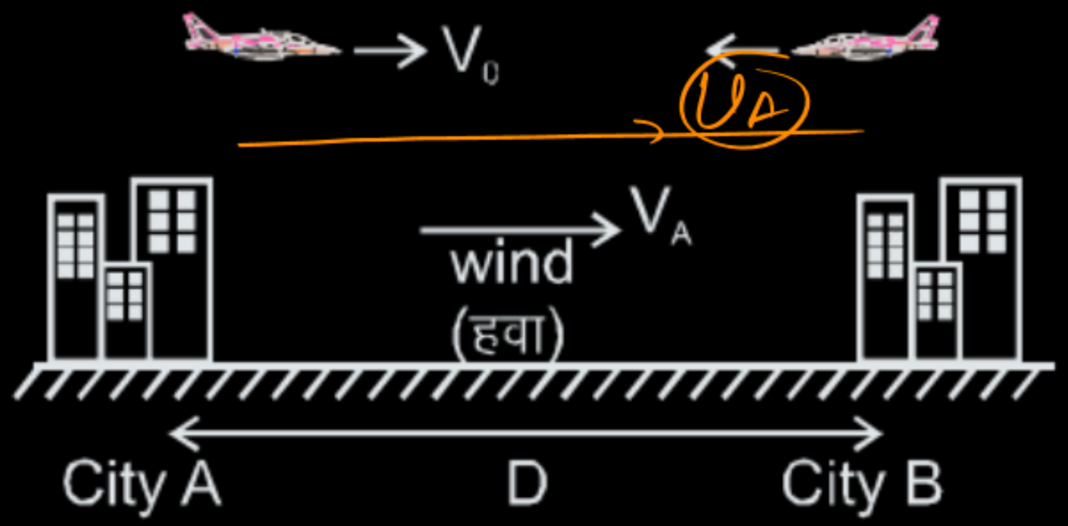


A
B
150m
50m/s = const

An airplane flies between two cities separated by a distance D. Assume the wind blows directly from one city to the other at a speed V_A (as shown) and the speed of the airplane is V_o relative to the air.

Find the time taken by the airplane to make a round trip between the two cities

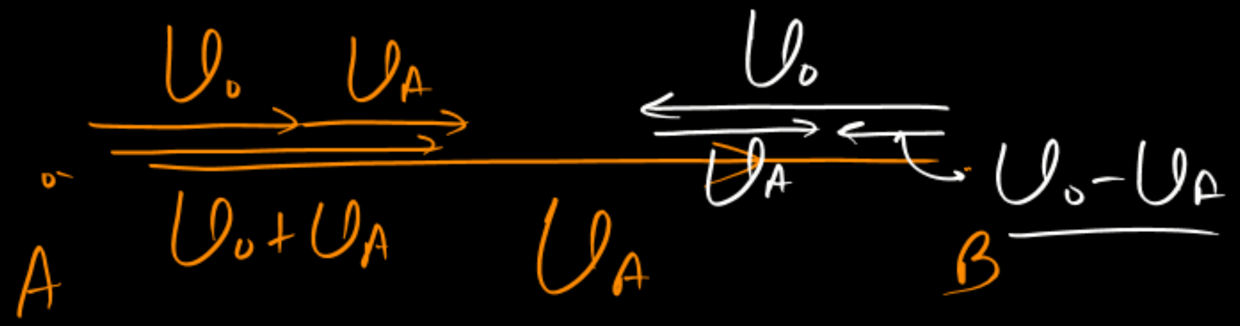
(that is, to fly from city A to city B and then back to City A) ?



(A) $\dfrac{2DV_o}{V_0^2 - V_A^2}$ ✓ (B) $\dfrac{DV_o}{V_0^2 - V_A^2}$ (C) $\dfrac{2DV_o}{V_0^2 + V_A^2}$ (D) $\dfrac{DV_o}{V_0^2 + V_A^2}$

$$\vec{U}_{AW} = \vec{U}_A - \vec{U}_W$$

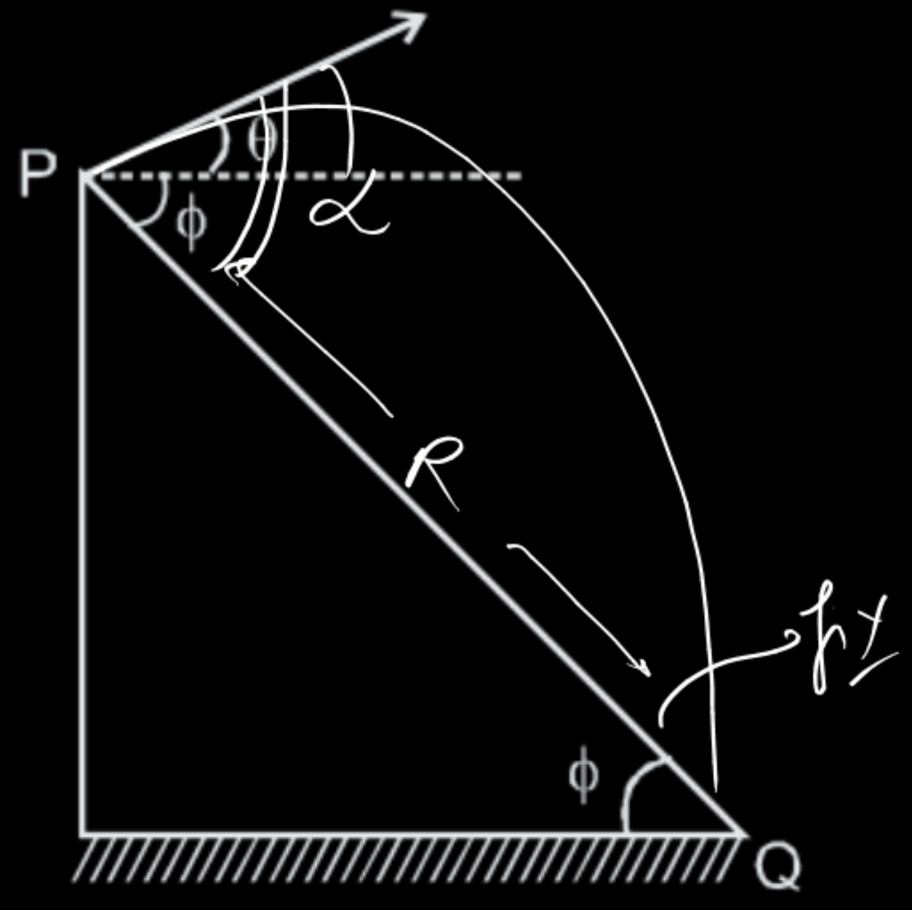
$$\vec{U}_A = \vec{U}_{AW} + \vec{U}_W$$



$$\frac{D}{U_0 + U_A} + \frac{D}{U_0 - U_A} = \frac{2DU_0}{U_0^2 - U_A^2}$$

A boy is standing at the peak of a hill which slopes downward uniformly at angle ϕ. At what angle 'θ' with the horizontal should he throw a ball, so that it has greatest range on slope PQ.

R_{max} $\quad a = const$



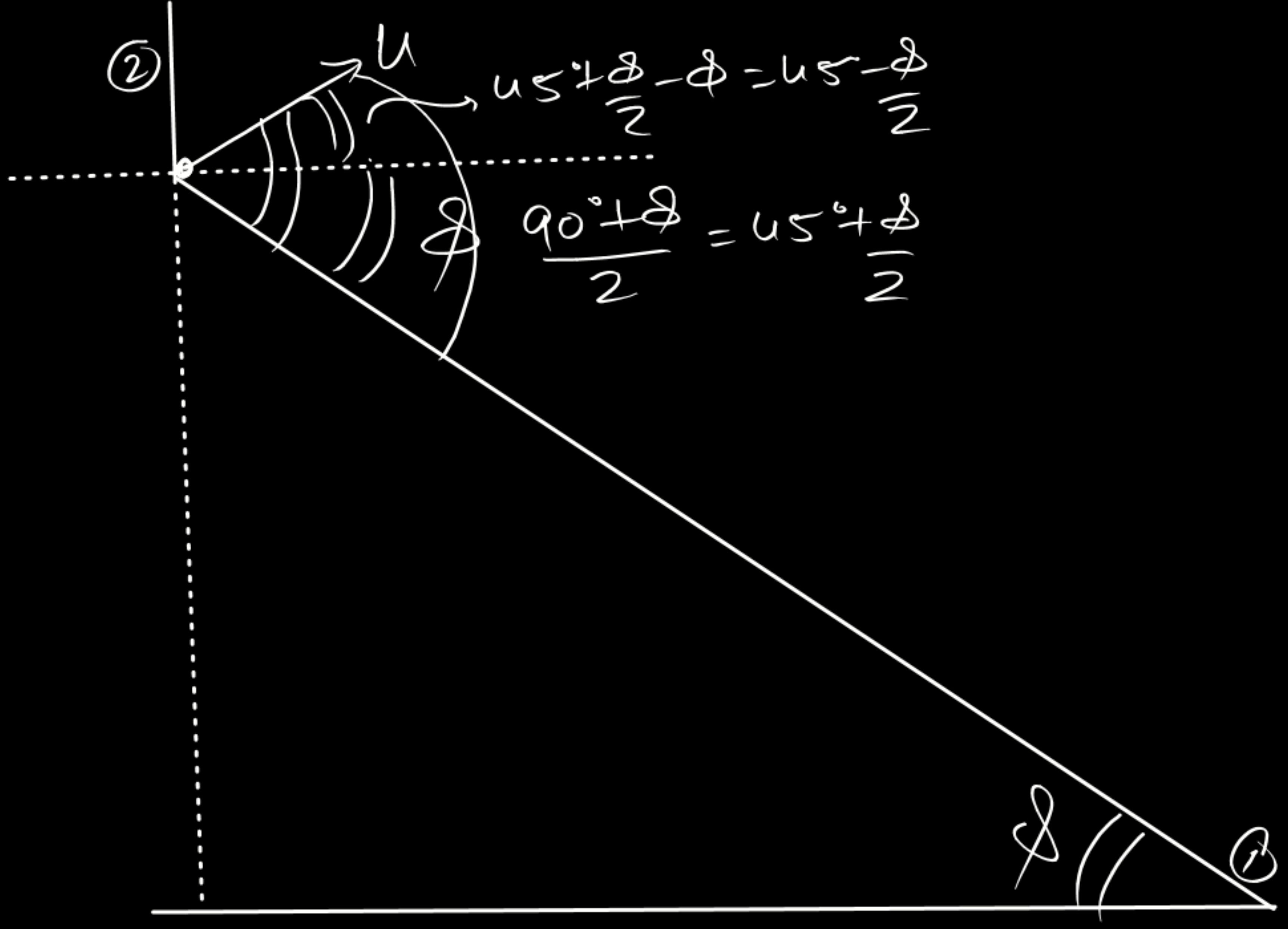
$$\frac{dR}{d\alpha} = 0$$

$$\alpha = 45^\circ + \frac{\phi}{2}$$

$$\alpha = 45^\circ + \frac{\phi}{2} - \phi$$

$$= 45^\circ - \frac{\phi}{2}$$

(A) $\frac{\phi}{2}$ (B) $90º + \frac{\phi}{2}$ (C) $45º - \frac{\phi}{2}$ (D) $45º + \frac{\phi}{2}$

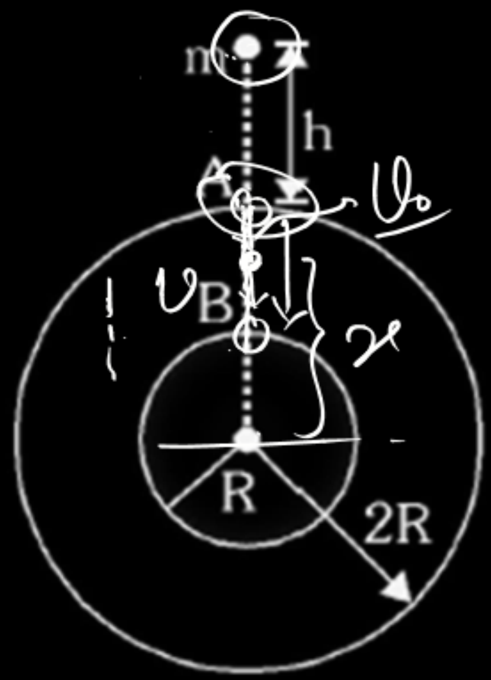


$$45° + \frac{\theta}{2} - \theta = 45 - \frac{\theta}{2}$$

$$\frac{90° + \theta}{2} = 45° + \frac{\theta}{2}$$

$\underline{90 + \theta}$

A solid sphere of mass M and radius R is surrounded by a sphericalshell of same mass M and radius 2R as shown. A small particle of mass m is released from rest from a height h (<<R) above the shell.There is a hole in the shell.



$$0 + m\left\{-\frac{GM}{2R} - \cancel{\frac{GM}{2R}}\right\} = \frac{1}{2}mv^2 + m\left\{-\frac{GM}{x} - \cancel{\frac{GM}{2R}}\right\}$$

$$v = -\frac{dx}{dt}$$

(A) time will it enter the hole at A is $2\sqrt{\frac{hR^2}{GM}}$ ✓

(B) time will it enter the hole at A is $\sqrt{\frac{hR^2}{GM}}$ ×

(C) time will it take to move from A to B is $< \frac{4R^2}{\sqrt{GMh}}$

(D) With what approximate speed will it collide at B is $\sqrt{\frac{GM}{R}}$

$$t < \frac{R}{v_0}$$

$$t < \frac{R}{\sqrt{\cancel{2} \times \frac{GM}{\cancel{2}R^2} \times h}}$$

$$\boxed{t < \frac{R^2}{\sqrt{GMh}}}$$

TMEC

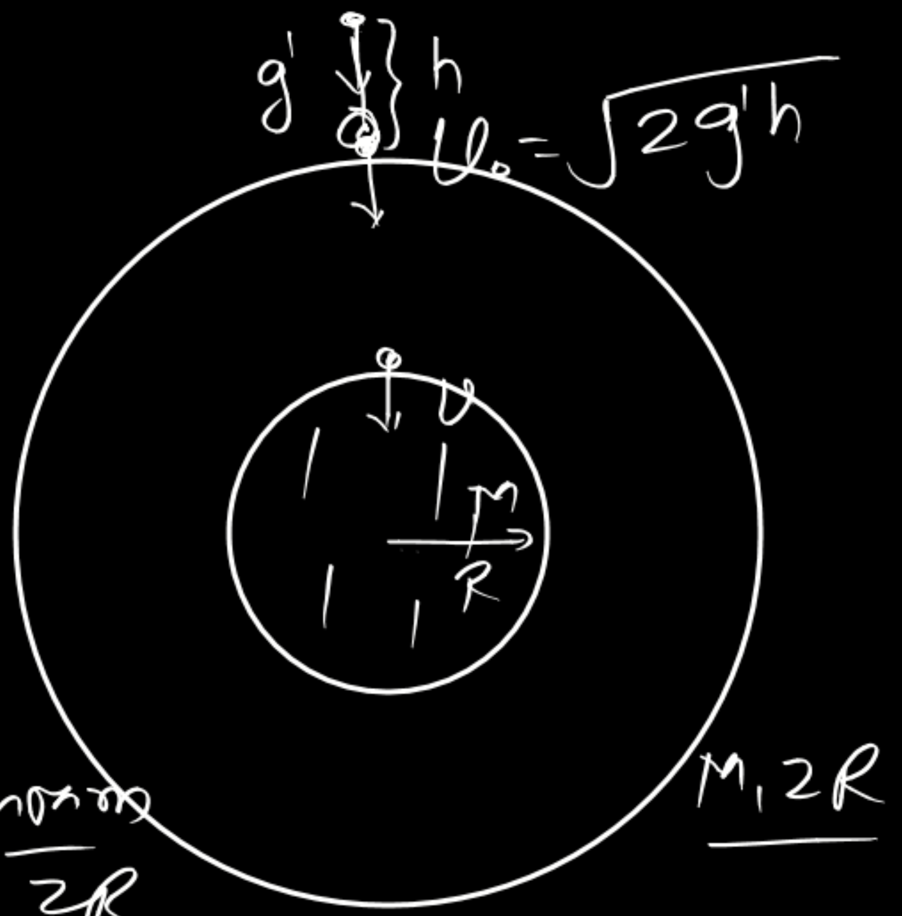
$$0 + m\left\{\cancel{-\frac{Gm}{2R}} - \frac{GM}{2R}\right\}$$

$$= \frac{1}{2} m v^2 + m\left\{\cancel{-\frac{Gm}{2R}} - \frac{GM}{R}\right\}$$

$$\frac{1}{2} m v^2 = \frac{GMm}{R} - \frac{GMm}{2R}$$

$$\frac{1}{2} m v^2 = \frac{GMm}{2R}$$

$$v = \sqrt{\frac{GM}{R}}$$



$$\frac{GMm}{(2R)^2} + \frac{GMm}{(2R)^2} = mg'$$

$$g' = \frac{GM}{2R^2}$$

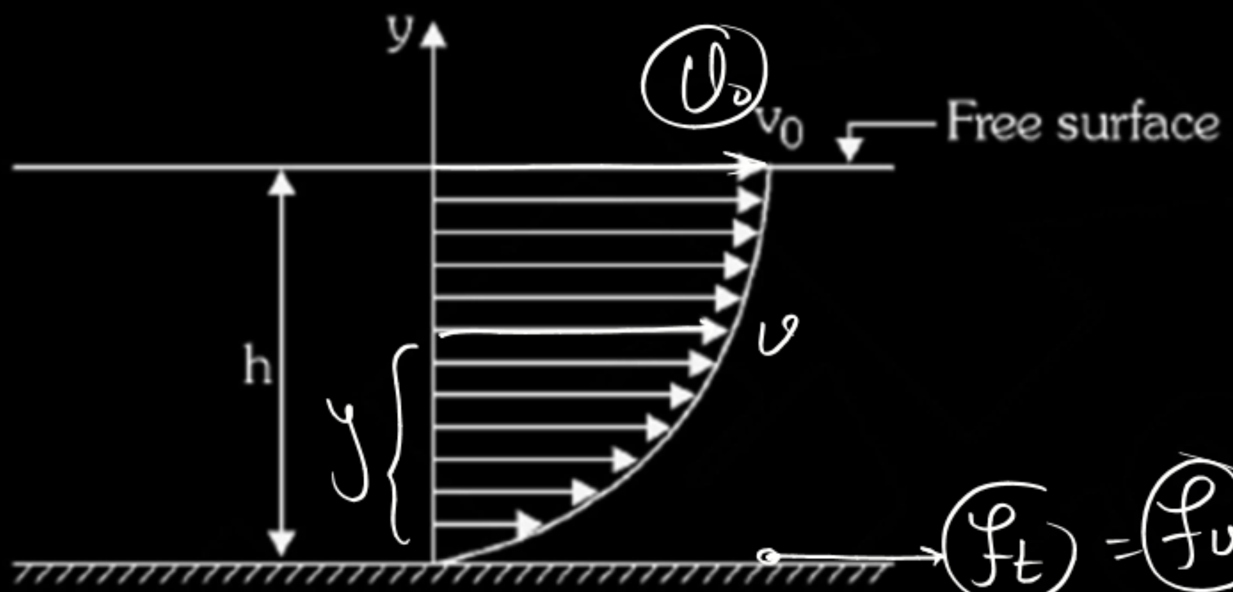
$$h = \frac{1}{2} g' t^2$$

$$t = \sqrt{\frac{2h \times 2R^2}{GM}} = \sqrt{\frac{4hR^2}{GM}}$$

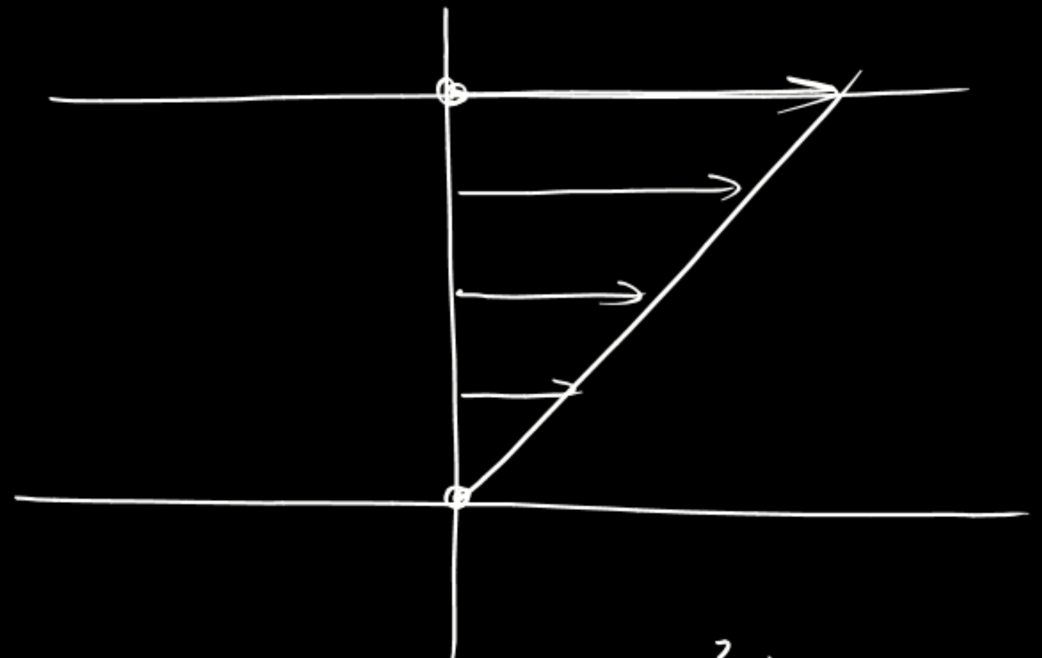
$$t = 2\sqrt{\frac{hR^2}{GM}}$$

A liquid is flowing a horizontal channel. The speed of flow (v) depends on height (y) from the floor as $v = v_o\left[2\left(\frac{y}{h}\right)-\left(\frac{y}{h}\right)^2\right]$

Where h is the height of liquid in the channel and v0 is the speed of the top layer. Coefficient of viscosity is η.



$\frac{f_t}{A} = f_v$



Choose the correct statement(s)

(A) The shear stress that the liquid exerts on the floor is $\frac{\eta V_0}{2h}$

(B) The shear stress that the liquid exerts on the floor is $\frac{2\eta V_0}{h}$

(C) The shear stress that the liquid exerts on the floor is $\left(\frac{\eta V_0}{3h}\right)$

(D) The expression for velocity gradient is $\left(\frac{2V_0}{h}-\frac{2V_0}{h^2}y\right)$

$v = v_0\left(\frac{2y}{h} - \frac{y^2}{h^2}\right)$

$$\frac{dv}{dy} = v_0\left\{\frac{2(1)}{h} - \frac{2y}{h^2}\right\} = \frac{2v_0}{h} - \frac{2v_0 y}{h^2}$$

$$F_v = \eta A \frac{dv}{dy}$$

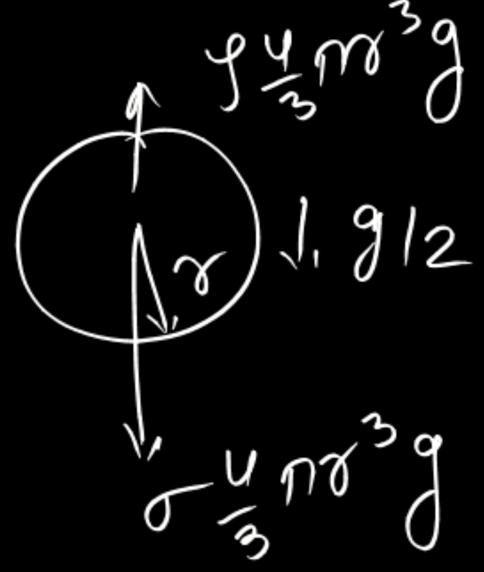
$$\frac{F_t}{A} = \frac{F_v}{A} = \eta \frac{dv}{dy}$$

$$= \eta \left\{ \frac{2v_0}{h} - \frac{2v_0 y}{h^2} \right\}$$

$\underline{y = 0}$

$$\frac{\eta 2v_0}{h}$$

Two balls of radii r and $\frac{r}{2}$ are released inside a deep water tank. Their initial accelerations are found to be $\frac{g}{2}$ and $\frac{g}{4}$ respectively. Coefficient of viscosity is given to be η. Take density of water ρ.



$\sigma \frac{4}{3}\cancel{\pi r^3} g - \rho \frac{4}{3}\cancel{\pi r^3} g$

$= \sigma \frac{4}{3}\cancel{\pi r^3} a$

$a = g - \frac{\rho}{\sigma} g = g/2$

$1 - \frac{\rho}{\sigma} = \frac{1}{2}$

$\frac{\rho}{\sigma} = \frac{1}{2}$

$\boxed{\sigma = 2\rho}$

Choose the correct statement(s):

terminal velo

(A) The velocity of smaller ball relative to large ball after a very long time the two balls are released is $\frac{11}{29}\left(\frac{r^2\rho g}{\eta}\right)$

(B) The velocity of smaller ball relative to large ball after a very long time the two balls are released Is $\frac{11}{54}\left(\frac{r^2\rho g}{\eta}\right)$

(C) The density of ball of radius r is 2ρ

(D) The density of ball of radius $\frac{r}{2}$ is $\frac{4\rho}{3}$

$\rho \frac{4}{3}\pi\left(\frac{r}{2}\right)^3 g$

$\downarrow g/4$

$\sigma' \frac{4}{3}\pi\left(\frac{r}{2}\right)^3 g$

$$v_A = \frac{2}{9}\frac{r^2(\rho)g}{\eta}$$

$$v_B = \frac{\cancel{2}}{9}\frac{\frac{r^2}{4\cancel{2}}\frac{\rho}{3}g}{\eta}$$

$$= \frac{1}{54}\frac{r^2\rho g}{\eta}$$

$$\sigma'\frac{4}{3}\cancel{\pi\left(\frac{r}{2}\right)^3}g - \rho\frac{4}{3}\cancel{\pi\left(\frac{r}{2}\right)^3}g = \sigma'\cancel{\frac{4}{3}\pi\left(\frac{r}{2}\right)^3}g/4$$

$$g - \frac{\rho}{\sigma'}g = g/4$$

$$1 - \frac{\rho}{\sigma'} = \frac{1}{4}$$

$$\frac{\rho}{\sigma'} = \frac{3}{4}$$

$$\boxed{\sigma' = \frac{4}{3}\rho}$$

$6\pi\eta r v$

$\rho\frac{4}{3}\pi r^3 g$

$\downarrow v$

$\sigma\frac{4}{3}\pi r^3 g$

$$\sigma\frac{4}{3}\cancel{\pi}r^3 g = \rho\frac{4}{3}\cancel{\pi}r^3 g + 6\cancel{\pi}\eta r v$$

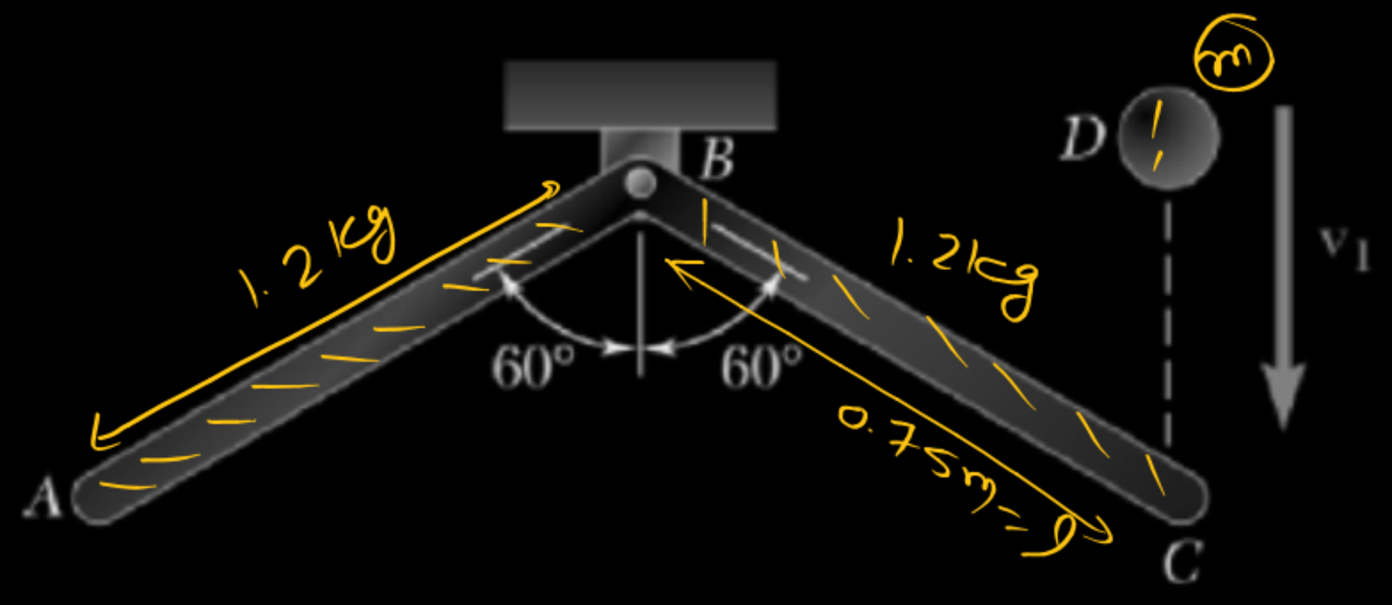
$$\boxed{\frac{2}{9}\frac{r^2 g(\sigma-\rho)}{\eta} = v}$$

$$\left(\frac{2}{9} - \frac{1}{54}\right)\frac{r^2\rho g}{\eta}$$

$$\frac{11}{54}\frac{r^2\rho g}{\eta}$$

$$P = Fv$$
$$= (6\pi\eta r v)v$$
$$= 6\pi\eta r v^2$$

Member ABC has a mass of 2.4 kg and is attached to a pin support at B. An 0.8 Kg sphere D strikes the end of member ABC with a vertical velocity v_1 of 3 m/s. Knowing that L = 0.75m and that the coefficient of restitution between the sphere and member ABC is 0.5, immediately after the impact –

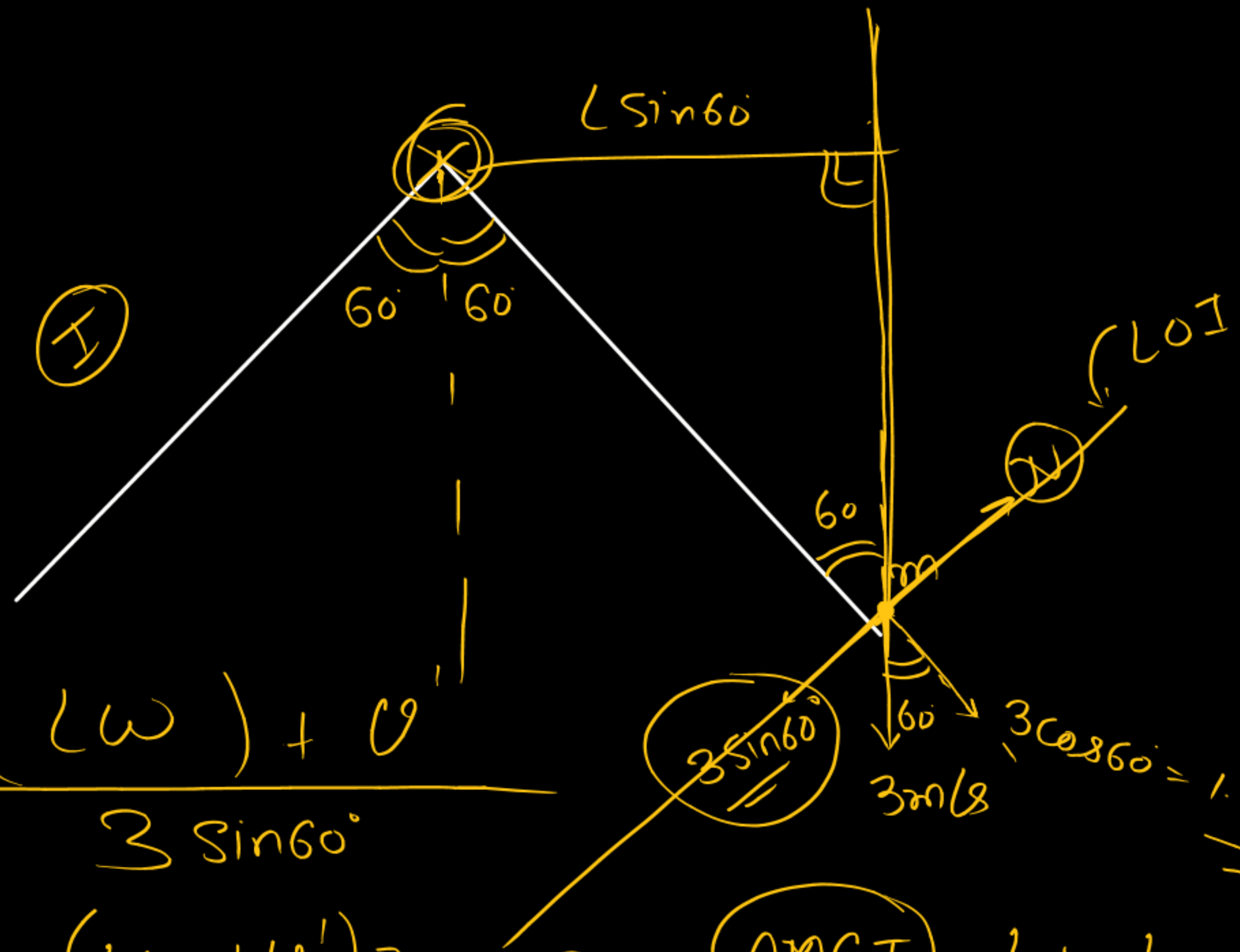


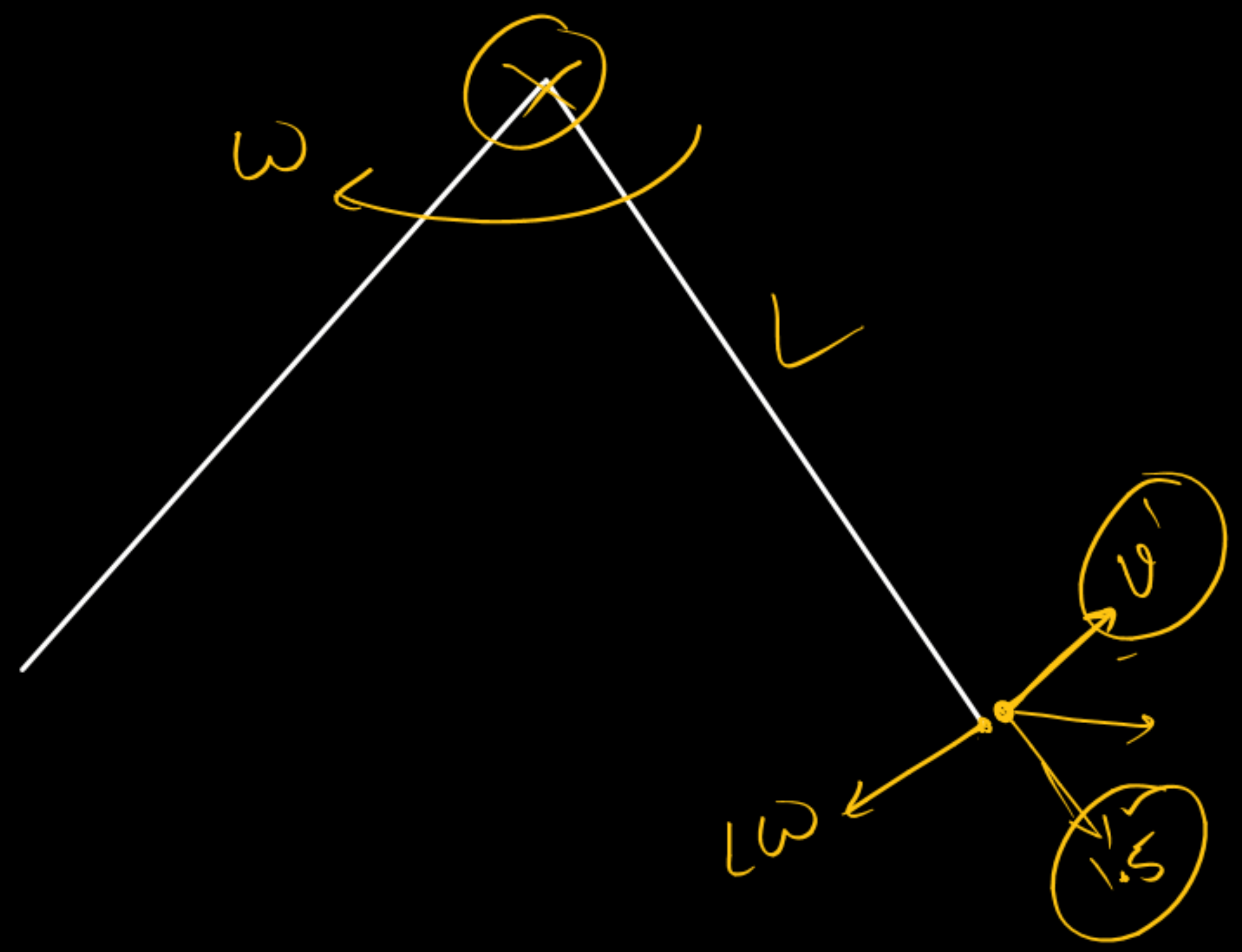
(A) the angular velocity of member ABC is 4.78 rad/s.

(B) the angular velocity of member ABC is 2.59 rad/s.

(C) speed of the sphere is 1.63 m/s.

(D) speed of the sphere is 2.28 m/s.





$$\frac{1}{2} = e = \frac{(L\omega) + v'}{3\sin 60^\circ}$$

$$\frac{1}{2} = \frac{(L\omega + v')2}{3\sqrt{3}} \quad \text{— (1)}$$

AMCT $\quad L_i = L_f$

$$0 + (m\,3L\sin 60)\otimes = I\omega\otimes + mv'L\odot$$

$$\frac{m3L\sqrt{3}}{2} = I\omega - mv'L$$

$$v' = 0.65$$

1.5 m/s = unchange

$$v = \sqrt{1.5^2 + 0.65^2} = 1.63 \text{ m/s}$$